FRONT WIPER MODULE (Continued)

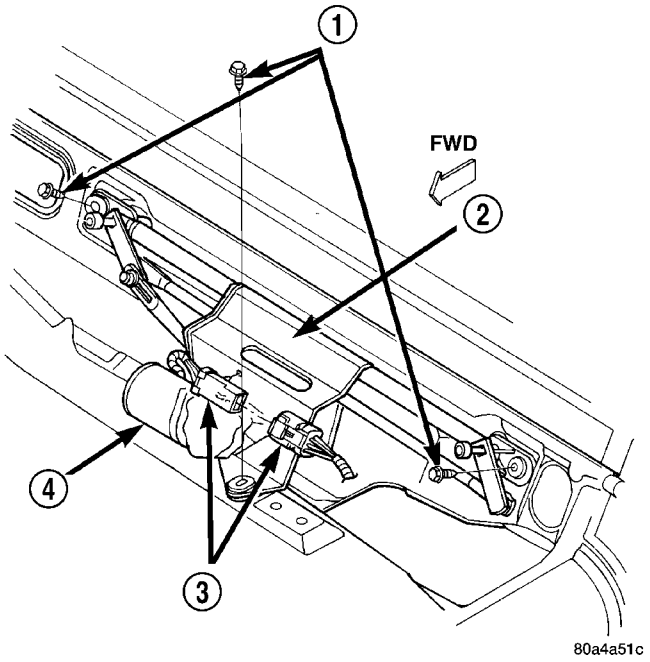

Fig. 17 Front Wiper Module

1 - SCREW (3)
2 - MODULE BRACKET
3 - WIRE HARNESS CONNECTOR
4 - WIPER MOTOR

- **Bracket** - The front wiper module bracket consists of a long tubular steel main member that has a stamped pivot bracket formation near each end where the two wiper pivots are secured. A stamped steel mounting plate for the wiper motor is secured with welds near the center of the main member. The bracket includes metal-sleeved rubber isolators at each of the three bracket mounting points.
- **Crank Arm** - The front wiper motor crank arm is a stamped steel unit with a slotted hole on the driven end that is secured to the wiper motor output shaft with a nut, and has a ball stud secured to the drive end.
- **Linkage** - The two front wiper linkage members are each constructed of stamped steel. A connecting link with a plastic socket-type bushing in each end is fit over the pivot ball studs to join the two pivots. The wiper drive link has a plastic socket-type bushing on each end. One end of the drive link is snap-fit over a second ball stud on the passenger side pivot crank arm, while the other end is snap-fit over the ball stud on the wiper motor crank arm.
- **Motor** - The front wiper motor is secured with three screws to the motor mounting plate near the center of the wiper module bracket and is protected by a rubber boot. The wiper motor output shaft passes through a hole in the module bracket, where a nut secures the wiper motor crank arm to the motor output shaft. The two-speed permanent magnet wiper motor features an integral transmission, an internal park switch, and an internal automatic resetting circuit breaker.
- **Pivots** - The two front wiper pivots are secured to the ends of the wiper module bracket. A crank arm extends from the lower end of each pivot shaft. The driver side pivot crank arm has a single ball stud secured to it, while the passenger side crank arm has two ball studs. The upper end of each pivot shaft where the wiper arms will be fastened each has an externally serrated drum-like driver secured to it.

The front wiper module for this model is serviced only as a complete unit. If any linkage component or the mounting bracket of the module is faulty or damaged, the entire front wiper module unit must be replaced. The front wiper motor and boot are available for service replacement as a unit only.

OPERATION

The front wiper module operation is controlled by the battery current inputs received by the wiper motor through the right multi-function switch on the steering column. The wiper motor is connected to the vehicle electrical system through a dedicated take out and wire harness connector of the body wire harness. The wiper motor speed is controlled by current flow to either the low speed or the high speed set of brushes. The park switch is a single pole, single throw, momentary switch within the wiper motor that is mechanically actuated by the wiper motor transmission components. The park switch alternately closes the wiper park switch sense circuit to ground or to battery current, depending upon the position of the wipers on the glass. This feature allows the motor to complete its current wipe cycle after the wiper system has been turned Off, and to park the wiper blades in the lowest portion of the wipe pattern. The automatic resetting circuit breaker protects the motor from overloads. The wiper motor crank arm, the two wiper linkage members, and the two wiper pivots mechanically convert the rotary output of the wiper motor to the back and forth wiping motion of the wiper arms and blades on the glass.

REMOVAL

(1) Remove the front wiper arms from the wiper pivots. (Refer to 8 - ELECTRICAL/FRONT WIPERS/WASHERS/FRONT WIPER ARM - REMOVAL).

(2) Unlatch, open and support the hood.

(3) Disconnect and isolate the battery negative cable.

(4) Remove the cowl plenum cover/grille panel from over the cowl plenum. (Refer to 23 - BODY/EXTERIOR/COWL GRILLE AND SCREEN - REMOVAL).